Solution:

Year	Season			
	I	II	III	IV
2001	118.4	260.0	379.4	70
2002	85.8	185.4	407.1	8.7
2003	129.8	336.5	403.1	12.0
2004	283.4	360.7	472.1	14.3
2005	231.7	308.5	828.8	15.9
Seasonal total	849.1	1451.1	2490.5	120.9
Seasonal average	169.82	290.22	498.1	24.18
Seasonal index	69	118	203	10

$$\text{Grand Average} = \frac{\text{Total of seasonal averages}}{4}$$

$$= \frac{169.82 + 290.22 + 498.1 + 24.18}{4}$$

$$= \frac{982.32}{4} = 245.58$$

$$\text{Seasonal Index} = \frac{\text{Seasonal average}}{\text{Grand average}} \times 100$$

$$\text{Seasonal Index for Season I} = \frac{169.82}{245.58} \times 100 = 69.15 \approx 69$$

$$\text{Seasonal Index for Season II} = \frac{290.22}{245.58} \times 100 = 118.18 \approx 118$$

$$\text{Seasonal Index for Season III} = \frac{498.1}{245.58} \times 100 = 202.83 \approx 203$$

$$\text{Seasonal Index for Season IV} = \frac{24.18}{245.58} \times 100 = 9.85 \approx 10$$

Example 7.10

Obtain the seasonal indices for the rain fall (in mm) data in India given in the following table.

Quarter \ Year	2009	2010	2011	2012
I	38.2	38.5	55	50.5
II	166.8	250.9	277.7	197
III	612.6	773.1	717.8	706.1
IV	72.2	153.1	65.8	101.1

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